

Run: overnight9h

Log-log loss curves

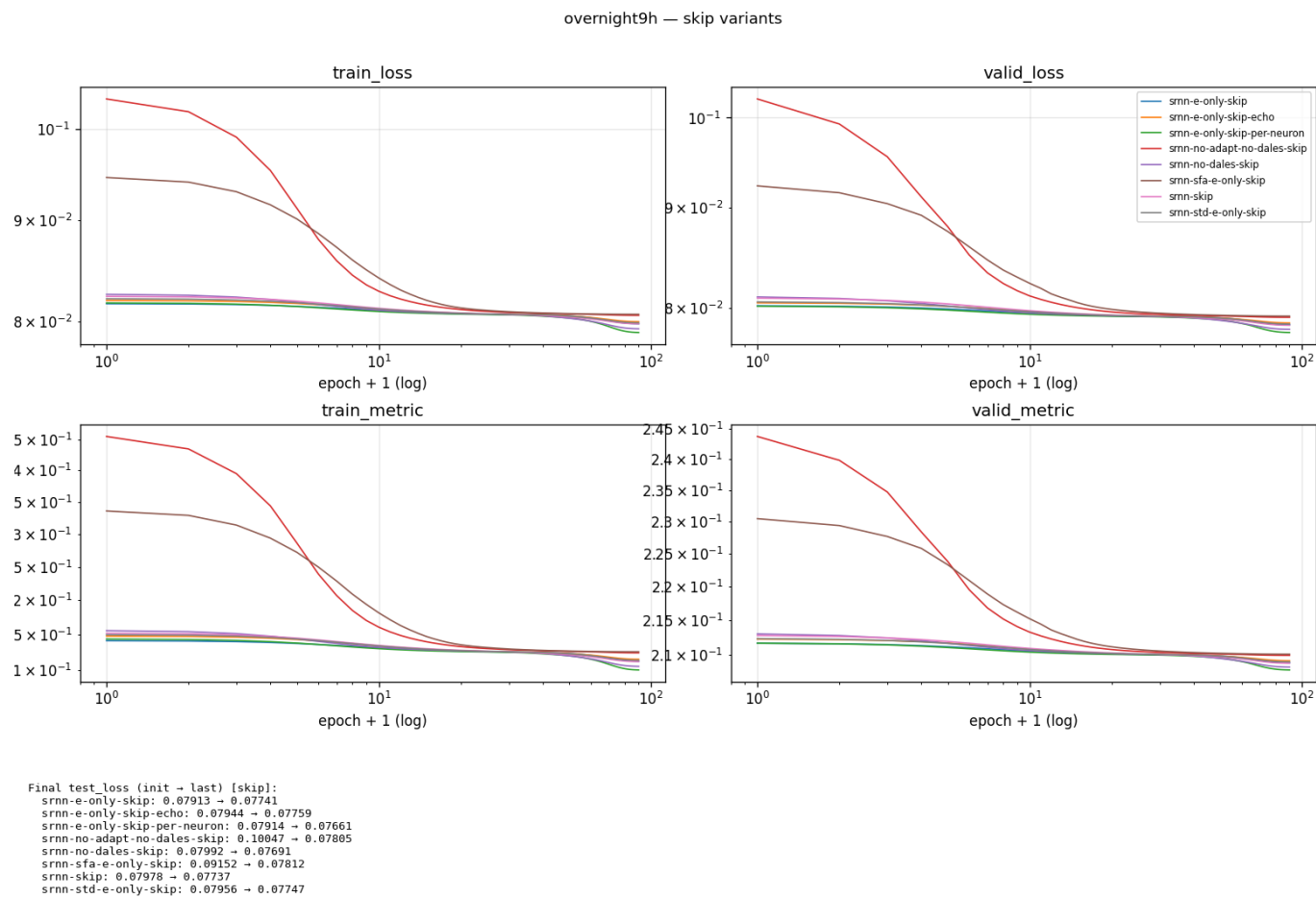


Figure 1: skip variants — log-log loss/metric

srnn-e-only-skip

overnight9h — effective time constants (srnn-e-only-skip)

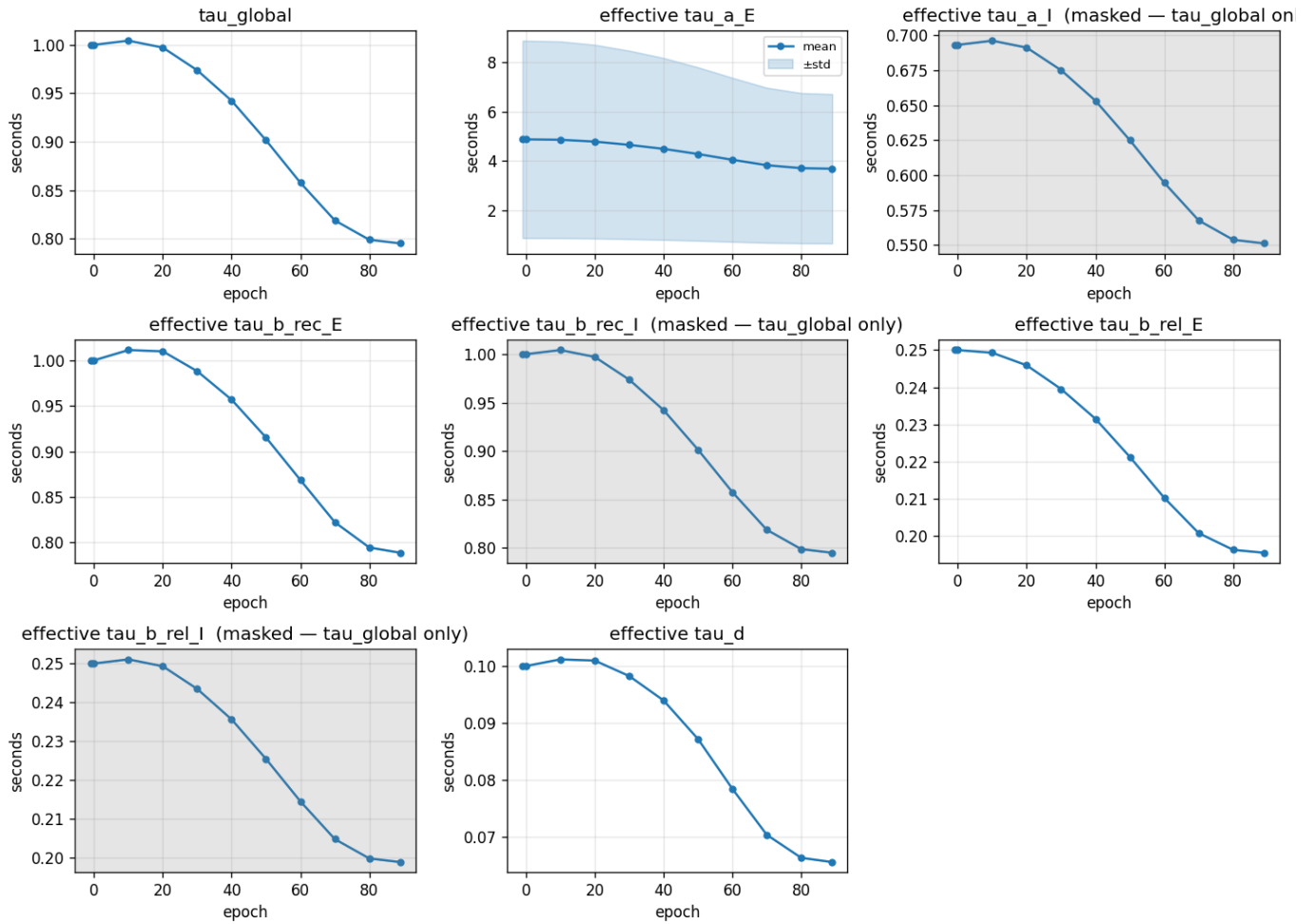
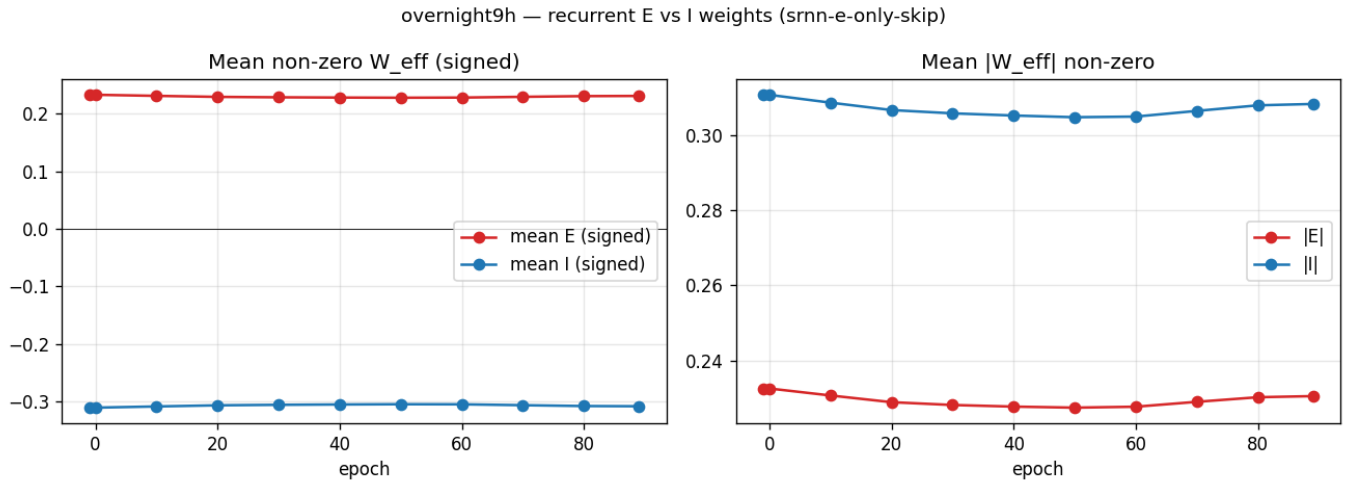


Figure 2: Tau evolution

Figure 3: W_{EI} evolution

srnn-e-only-skip — parameter table

# overnight9h - variant srnn-e-only-skip (k=0)					
# Effective values (post-transform). Means/stds computed over the indicated dimension.					
effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.79508	-	-	scalar
tau_d (s)	+0.1	+0.065561	0	0	300
tau_a_E (s)	+4.8777	+3.6887	3.996	3.022	450
tau_a_I (s)	+0.69315	+0.55111	0	0	450
tau_b_rec_E (s)	+1	+0.78867	0	0	150
tau_b_rel_E (s)	+0.25	+0.19556	0	0	150
tau_b_rec_I (s)	+1	+0.79508	0	0	150
tau_b_rel_I (s)	+0.25	+0.19877	0	0	150
W_{eff} (E src, signed)	+0.23267	+0.23063	0.07641	0.076	15048
W_{eff} (I src, signed)	-0.31062	-0.30818	0.07724	0.07696	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.23063	0.07641	0.076	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30818	0.07724	0.07696	14977
$W_{\text{in_eff}}$ (input neurons)	-0.0003386	+0.0037668	0.1012	0.1113	6675
c_E (SFA coupling)	+0.05	+0.051138	0	0	450
c_0_E (SFA offset)	+0	+0.021846	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36556	0	0	300
readout_weight	-0.00041422	-0.0076553	0.06568	0.0598	6675
readout_bias	+0	+0.012703	0	0.01728	89
ic.ic (per-variant init)	+0.15932	+0.17372	0.4111	0.4079	1500

srnn-e-only-skip-per-neuron

overnight9h — effective time constants (srnn-e-only-skip-per-neuron)

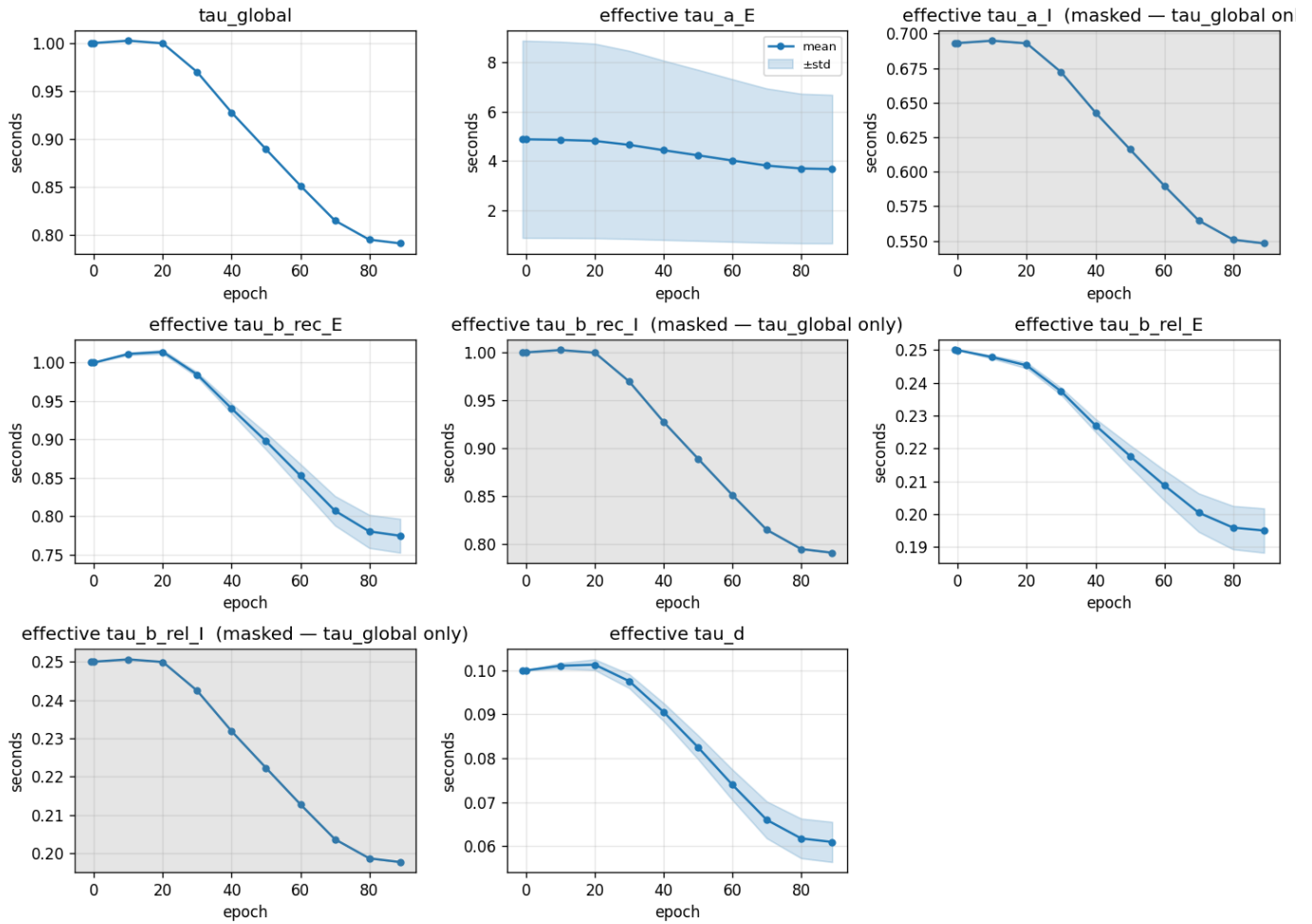


Figure 4: Tau evolution

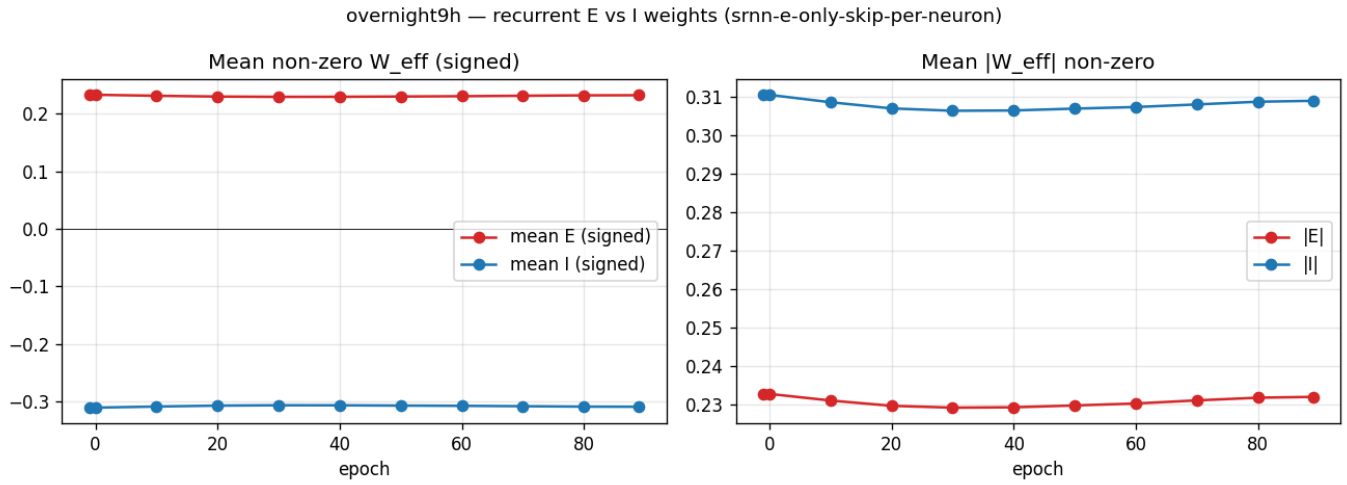


Figure 5: W_EI evolution

srnn-e-only-skip-per-neuron — parameter table

# overnight9h - variant srnn-e-only-skip-per-neuron (k=1)					
# Effective values (post-transform). Means/stds computed over the indicated dimension.					
effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.79081	-	-	scalar
tau_d (s)	+0.1	+0.060951	0	0.00457	300
tau_a_E (s)	+4.8777	+3.6689	3.996	3.01	450
tau_a_I (s)	+0.69315	+0.54815	0	0	450
tau_b_rec_E (s)	+1	+0.77469	0	0.02205	150
tau_b_rel_E (s)	+0.25	+0.19497	0	0.006761	150
tau_b_rec_I (s)	+1	+0.79081	0	0	150
tau_b_rel_I (s)	+0.25	+0.1977	0	0	150
W_eff (E src, signed)	+0.23267	+0.23188	0.07641	0.07641	15048
W_eff (I src, signed)	-0.31062	-0.30904	0.07724	0.07736	14977
W_eff (E src)	+0.23267	+0.23188	0.07641	0.07641	15048
W_eff (I src)	+0.31062	+0.30904	0.07724	0.07736	14977
W_in_eff (input neurons)	-0.00084237	+0.00059068	0.1008	0.08985	6675
c_E (SFA coupling)	+0.05	+0.051772	0	0.001311	450
c_0_E (SFA offset)	+0	+0.026169	0	0.0242	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.35272	0	0.04392	300
readout_weight	-0.0016154	-0.0090911	0.06624	0.06309	6675
readout_bias	+0	+0.027875	0	0.03333	89
ic.ic (per-variant init)	+0.15938	+0.16418	0.4127	0.4249	1500

srnn-e-only-skip-echo

overnight9h — effective time constants (srnn-e-only-skip-echo)

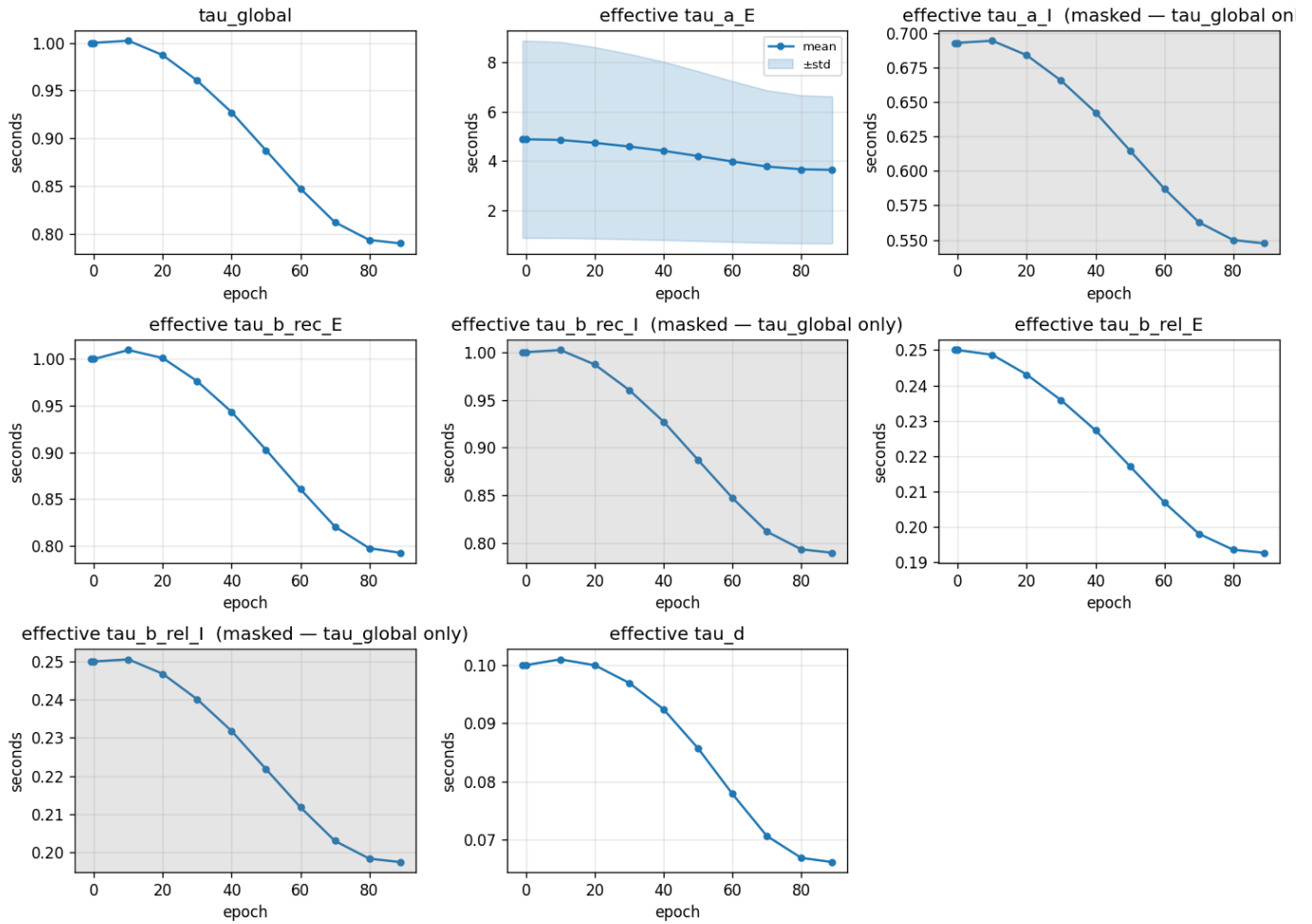
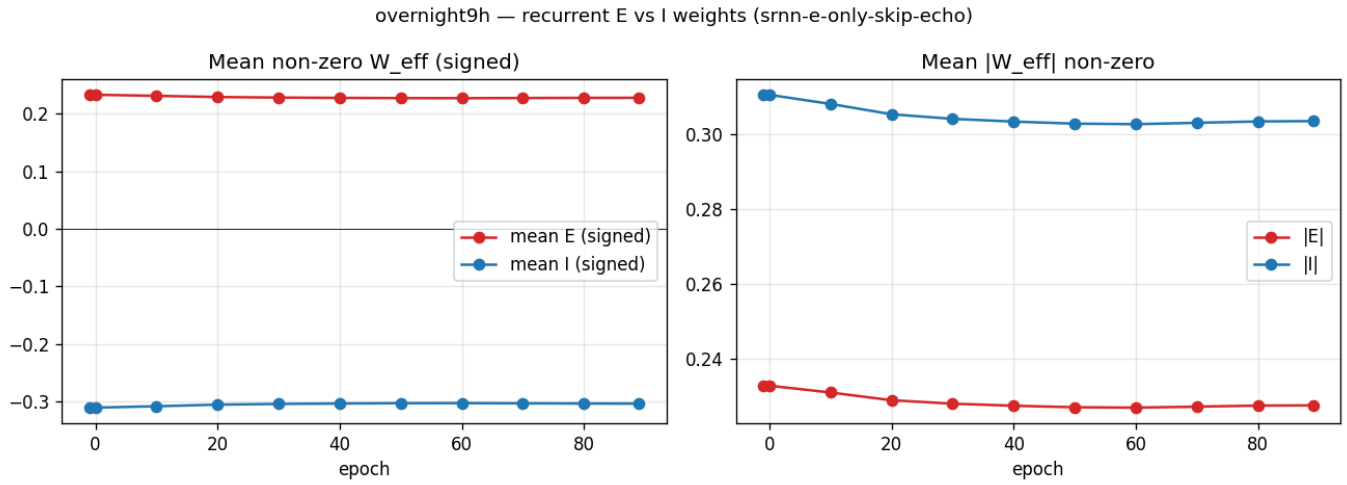


Figure 6: Tau evolution

Figure 7: W_{EI} evolution

srnn-e-only-skip-echo — parameter table

# overnight9h - variant srnn-e-only-skip-echo (k=2)					
# Effective values (post-transform). Means/stds computed over the indicated dimension.					
effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.7898	-	-	scalar
tau_d (s)	+0.1	+0.066188	0	0	300
tau_a_E (s)	+4.8777	+3.637	3.996	2.979	450
tau_a_I (s)	+0.69315	+0.54745	0	0	450
tau_b_rec_E (s)	+1	+0.79227	0	0	150
tau_b_rel_E (s)	+0.25	+0.19272	0	0	150
tau_b_rec_I (s)	+1	+0.7898	0	0	150
tau_b_rel_I (s)	+0.25	+0.19745	0	0	150
W_{eff} (E src, signed)	+0.23267	+0.22739	0.07641	0.07468	15048
W_{eff} (I src, signed)	-0.31062	-0.30357	0.07724	0.07549	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.22739	0.07641	0.07468	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30357	0.07724	0.07549	14977
$W_{\text{in_eff}}$ (input neurons)	-0.00015382	+0.0030492	0.09934	0.1059	6675
c_E (SFA coupling)	+0.05	+0.051541	0	0	450
c_0_E (SFA offset)	+0	+0.027732	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.37416	0	0	300
readout_weight	-0.00011373	-0.0096042	0.06687	0.05935	6675
readout_bias	+0	+0.010302	0	0.01316	89
ic.ic (per-variant init)	+0.15962	+0.17629	0.4124	0.4045	1500

srnn-no-dales-skip

overnight9h — effective time constants (srnn-no-dales-skip)

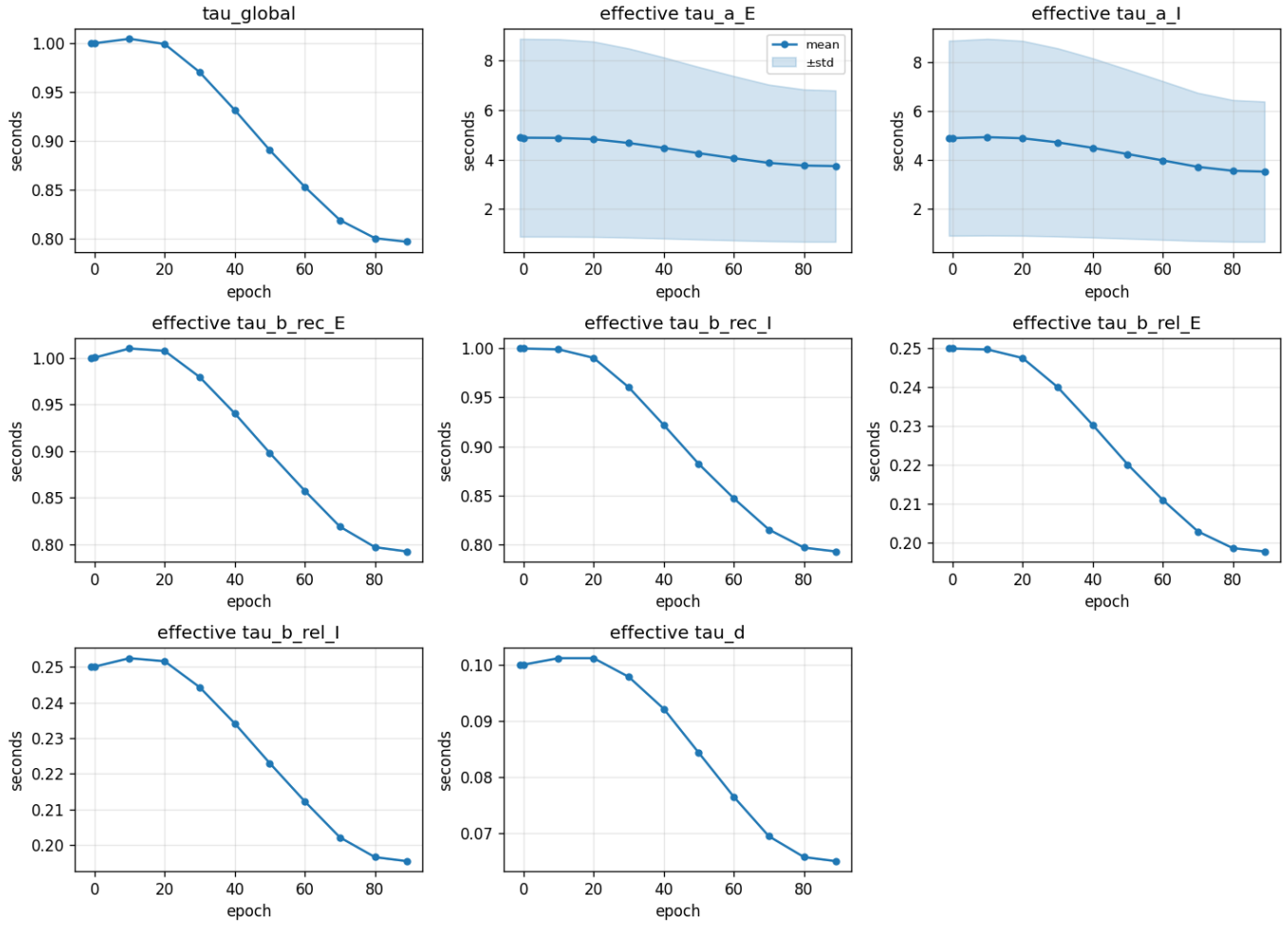
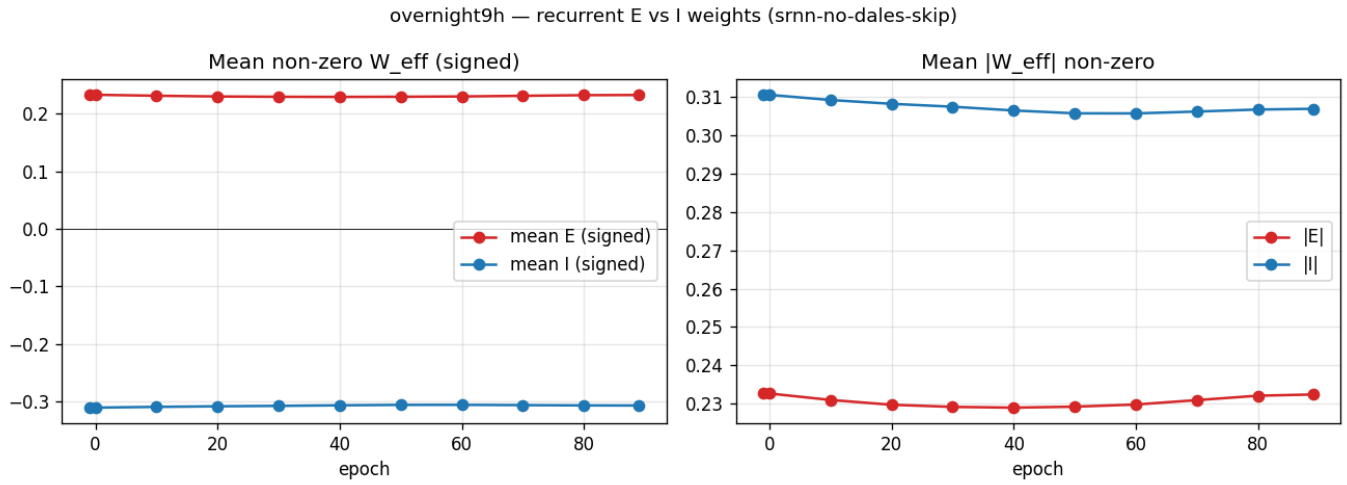


Figure 8: Tau evolution

Figure 9: W_{EI} evolution

srnn-no-dales-skip — parameter table

overnight9h - variant srnn-no-dales-skip (k=3)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.797	-	-	scalar
tau_d (s)	+0.1	+0.06505	0	0	300
tau_a_E (s)	+4.8777	+3.7314	3.996	3.057	450
tau_a_I (s)	+4.8777	+3.5061	3.996	2.872	450
tau_b_rec_E (s)	+1	+0.79268	0	0	150
tau_b_rel_E (s)	+0.25	+0.19763	0	0	150
tau_b_rec_I (s)	+1	+0.79265	0	0	150
tau_b_rel_I (s)	+0.25	+0.19557	0	0	150
W_{eff} (E src, signed)	+0.2326	+0.23221	0.07663	0.08022	15048
W_{eff} (I src, signed)	-0.31062	-0.30686	0.07727	0.08275	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.23239	0.07641	0.07972	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30696	0.07724	0.0824	14977
$W_{\text{in_eff}}$ (input neurons)	-0.0011621	+0.0049221	0.09962	0.1037	6675
c_E (SFA coupling)	+0.05	+0.050413	0	0	450
c_0_E (SFA offset)	+0	+0.0083969	0	0	450
c_I (SFA coupling)	+0.05	+0.05011	0	0	450
c_0_I (SFA offset)	+0	-0.012404	0	0	450
a_0 (threshold)	+0.35	+0.35191	0	0	300
readout_weight	-0.0016591	-0.012663	0.06668	0.06659	6675
readout_bias	+0	+0.020652	0	0.02351	89
ic.ic (per-variant init)	+0.16747	+0.16416	0.3627	0.3999	1500

srnn-no-adapt-no-dales-skip

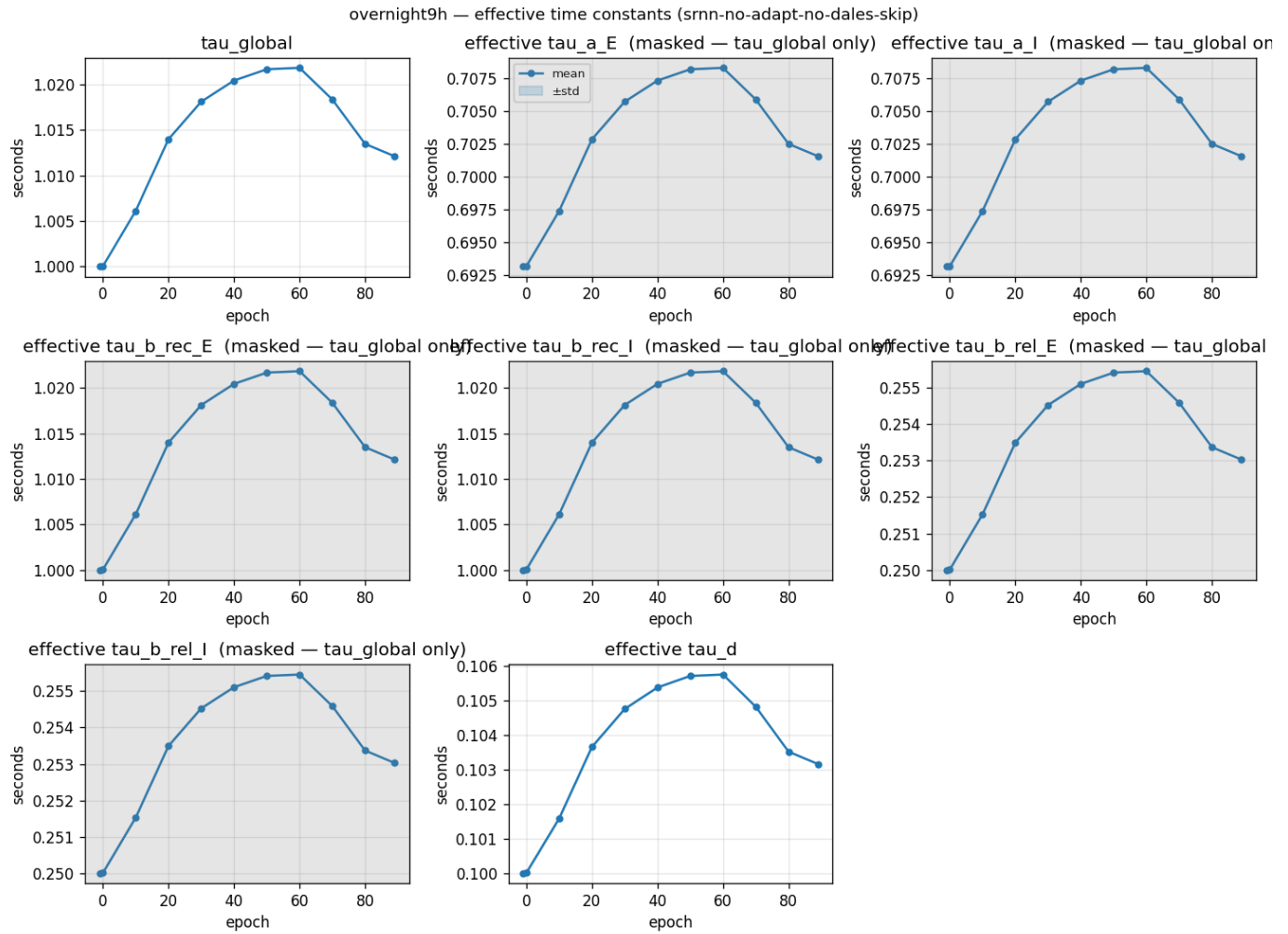
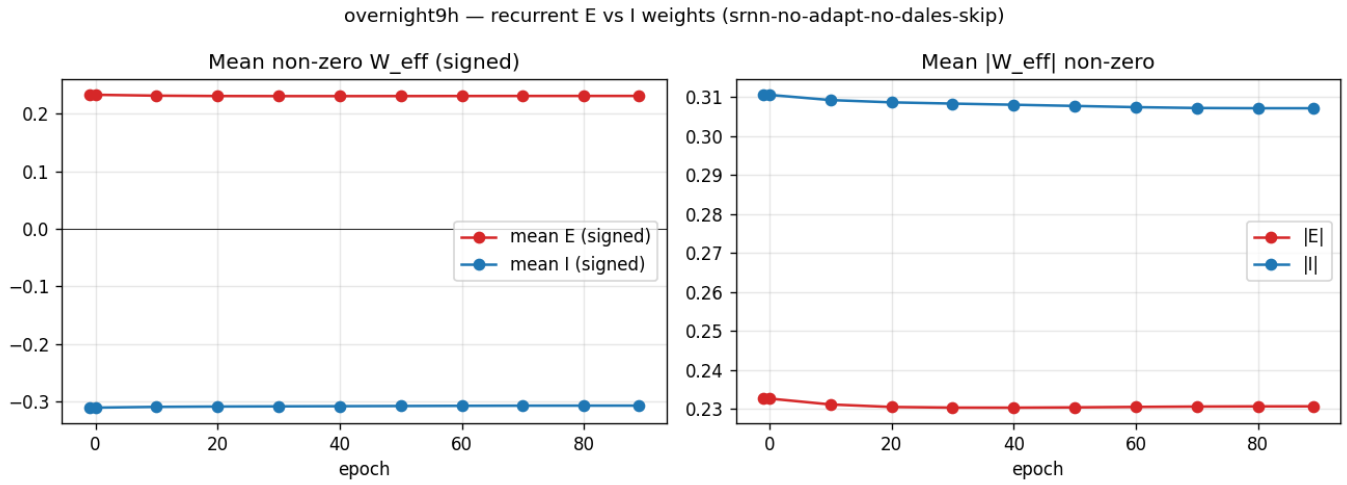


Figure 10: Tau evolution

Figure 11: W_{EI} evolution

srnn-no-adapt-no-dales-skip — parameter table

overnight9h - variant srnn-no-adapt-no-dales-skip (k=4)

Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+1.0121	-	-	scalar
tau_d (s)	+0.1	+0.10317	0	0	300
tau_a_E (s)	+0.69315	+0.70155	0	0	450
tau_a_I (s)	+0.69315	+0.70155	0	0	450
tau_b_rec_E (s)	+1	+1.0121	0	0	150
tau_b_rel_E (s)	+0.25	+0.25303	0	0	150
tau_b_rec_I (s)	+1	+1.0121	0	0	150
tau_b_rel_I (s)	+0.25	+0.25303	0	0	150
W_{eff} (E src, signed)	+0.2326	+0.23059	0.07663	0.07752	15048
W_{eff} (I src, signed)	-0.31062	-0.30708	0.07727	0.08045	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.23066	0.07641	0.07732	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30716	0.07724	0.08013	14977
$W_{\text{in_eff}}$ (input neurons)	+0.00019944	+0.0007215	0.09994	0.09496	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.36811	0	0	300
readout_weight	+0.00079717	+0.00042102	0.06573	0.06185	6675
readout_bias	+0	+0.0006623	0	0.002357	89
ic.ic (per-variant init)	+0.14435	+0.16403	0.4566	0.4252	1500

srnn-skip

overnight9h — effective time constants (srnn-skip)

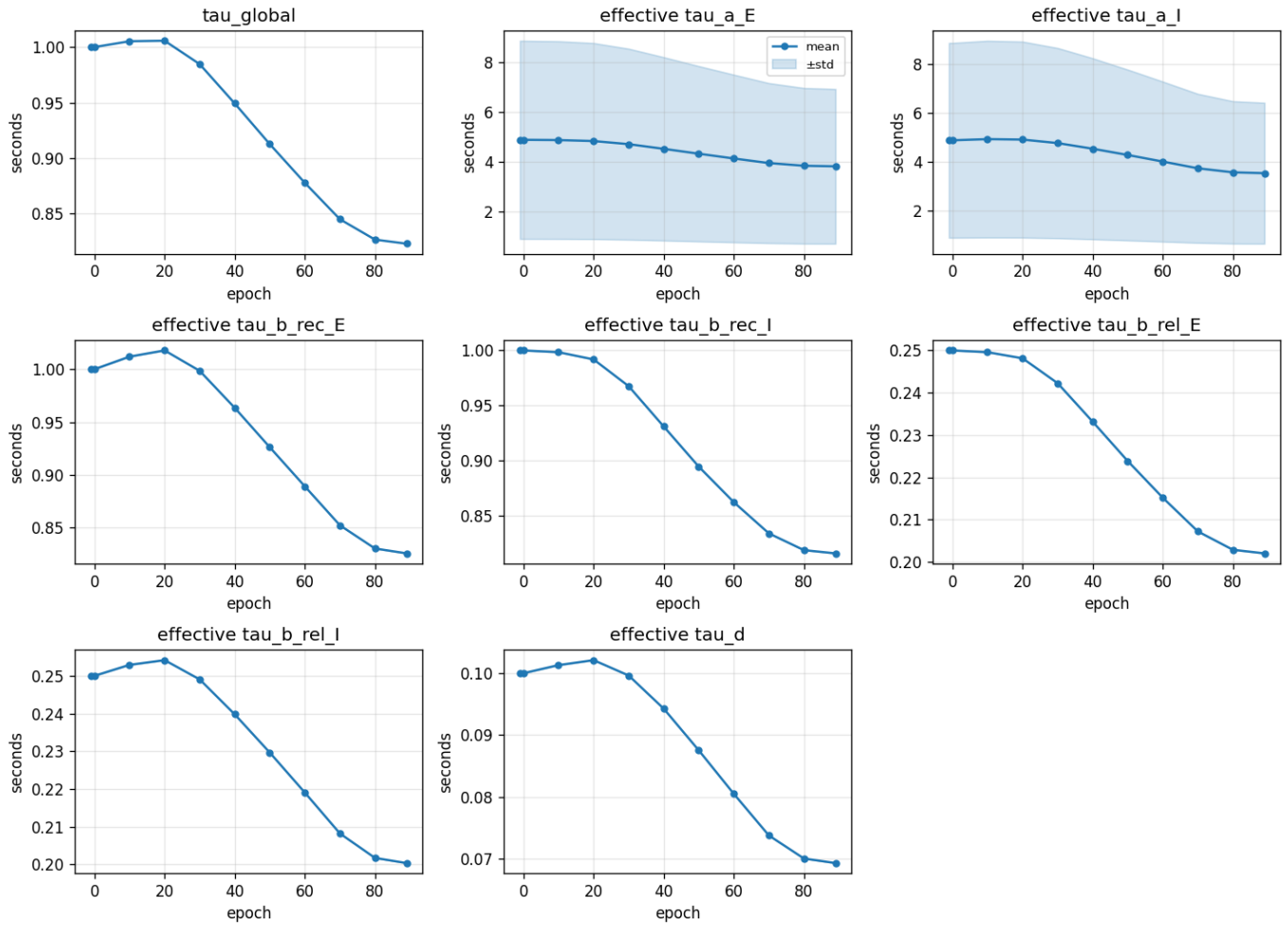


Figure 12: Tau evolution

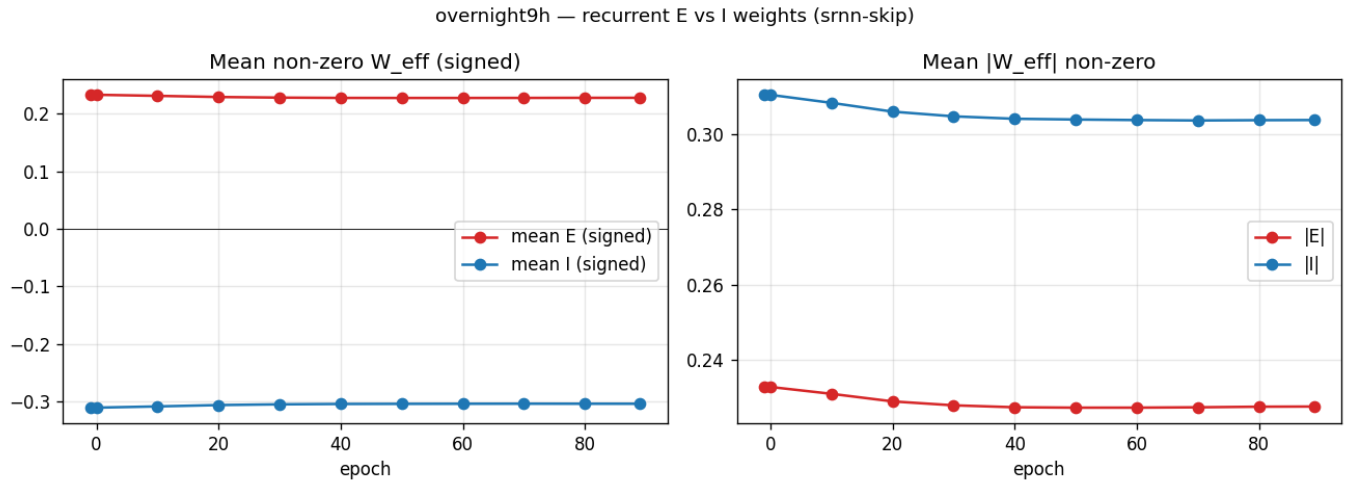


Figure 13: W_EI evolution

srnn-skip — parameter table

overnight9h - variant srnn-skip (k=5)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.82291	-	-	scalar
tau_d (s)	+0.1	+0.069289	0	0	300
tau_a_E (s)	+4.8777	+3.8078	3.996	3.119	450
tau_a_I (s)	+4.8777	+3.5291	3.996	2.891	450
tau_b_rec_E (s)	+1	+0.8256	0	0	150
tau_b_rel_E (s)	+0.25	+0.20204	0	0	150
tau_b_rec_I (s)	+1	+0.8156	0	0	150
tau_b_rel_I (s)	+0.25	+0.20033	0	0	150
W_eff (E src, signed)	+0.23267	+0.22743	0.07641	0.07487	15048
W_eff (I src, signed)	-0.31062	-0.30388	0.07724	0.07581	14977
W_eff (E src)	+0.23267	+0.22743	0.07641	0.07487	15048
W_eff (I src)	+0.31062	+0.30388	0.07724	0.07581	14977
W_in_eff (input neurons)	+0.0005819	+0.00453	0.09954	0.1013	6675
c_E (SFA coupling)	+0.05	+0.051033	0	0	450
c_0_E (SFA offset)	+0	+0.021421	0	0	450
c_I (SFA coupling)	+0.05	+0.049782	0	0	450
c_0_I (SFA offset)	+0	-0.024307	0	0	450
a_0 (threshold)	+0.35	+0.36655	0	0	300
readout_weight	-0.00025143	-0.0087016	0.06686	0.06239	6675
readout_bias	+0	+0.011804	0	0.01621	89
ic.ic (per-variant init)	+0.17068	+0.16826	0.3591	0.3885	1500

srnn-sfa-e-only-skip

overnight9h — effective time constants (srnn-sfa-e-only-skip)

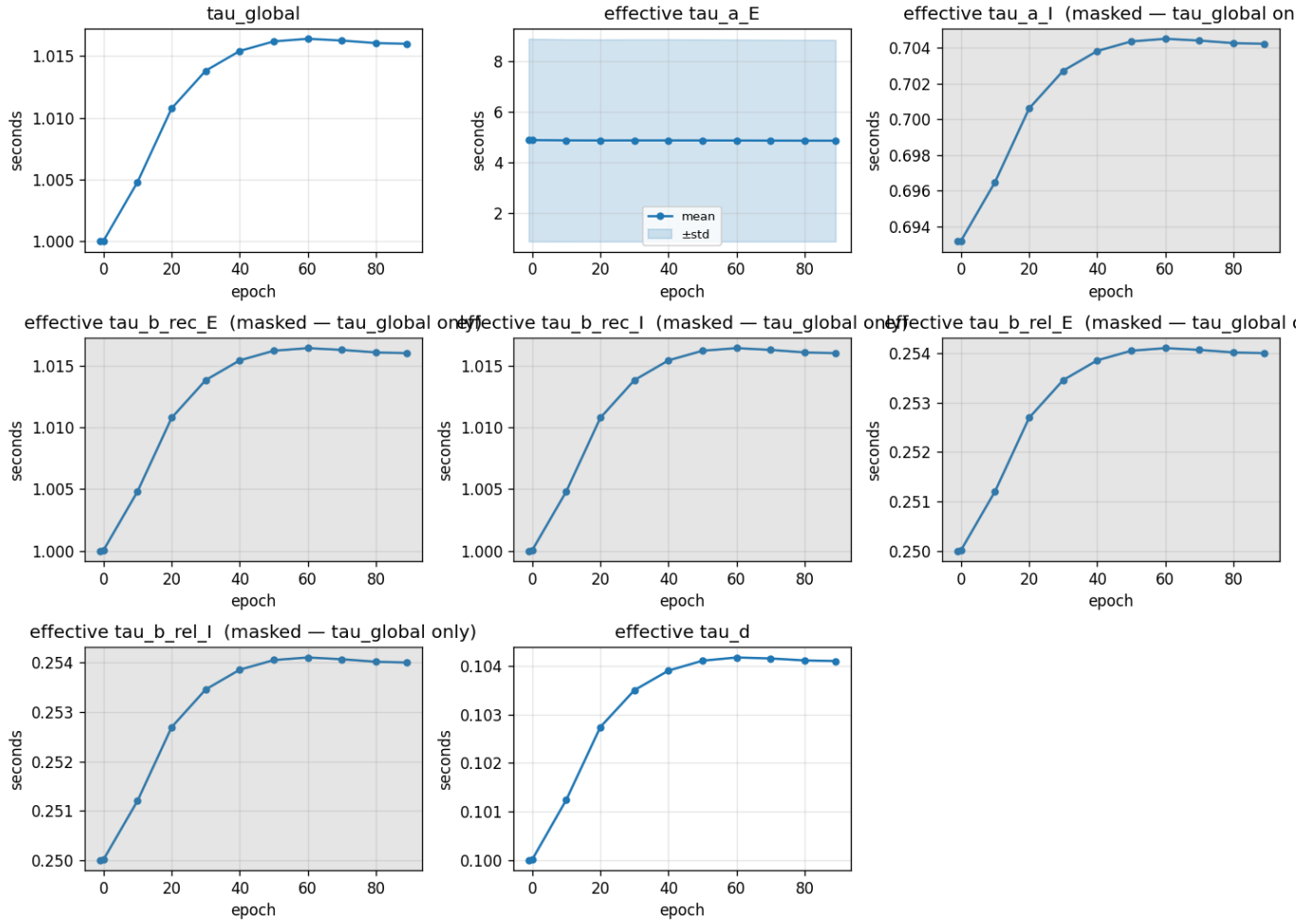
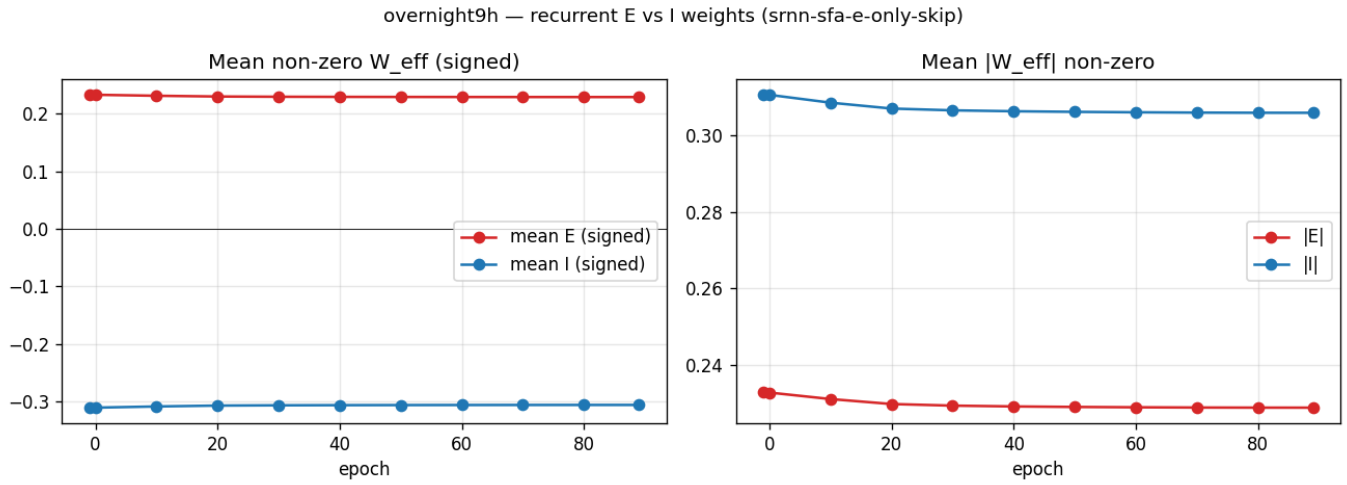


Figure 14: Tau evolution

Figure 15: W_{EI} evolution

srnn-sfa-e-only-skip — parameter table

overnight9h - variant srnn-sfa-e-only-skip (k=6)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+1.016	-	-	scalar
tau_d (s)	+0.1	+0.1041	0	0	300
tau_a_E (s)	+4.8777	+4.8536	3.996	3.976	450
tau_a_I (s)	+0.69315	+0.70424	0	0	450
tau_b_rec_E (s)	+1	+1.016	0	0	150
tau_b_rel_E (s)	+0.25	+0.254	0	0	150
tau_b_rec_I (s)	+1	+1.016	0	0	150
tau_b_rel_I (s)	+0.25	+0.254	0	0	150
W_{eff} (E src, signed)	+0.23267	+0.2287	0.07641	0.07517	15048
W_{eff} (I src, signed)	-0.31062	-0.3059	0.07724	0.07613	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.2287	0.07641	0.07517	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.3059	0.07724	0.07613	14977
$W_{\text{in_eff}}$ (input neurons)	-0.00083129	-0.00069762	0.09974	0.09659	6675
c_E (SFA coupling)	+0.05	+0.050765	0	0	450
c_0_E (SFA offset)	+0	+0.023817	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.37963	0	0	300
readout_weight	+0.001119	-0.00030331	0.06703	0.06107	6675
readout_bias	+0	-0.00045903	0	0.00227	89
ic.ic (per-variant init)	+0.16079	+0.17861	0.4384	0.4194	1500

srnn-std-e-only-skip

overnight9h — effective time constants (srnn-std-e-only-skip)

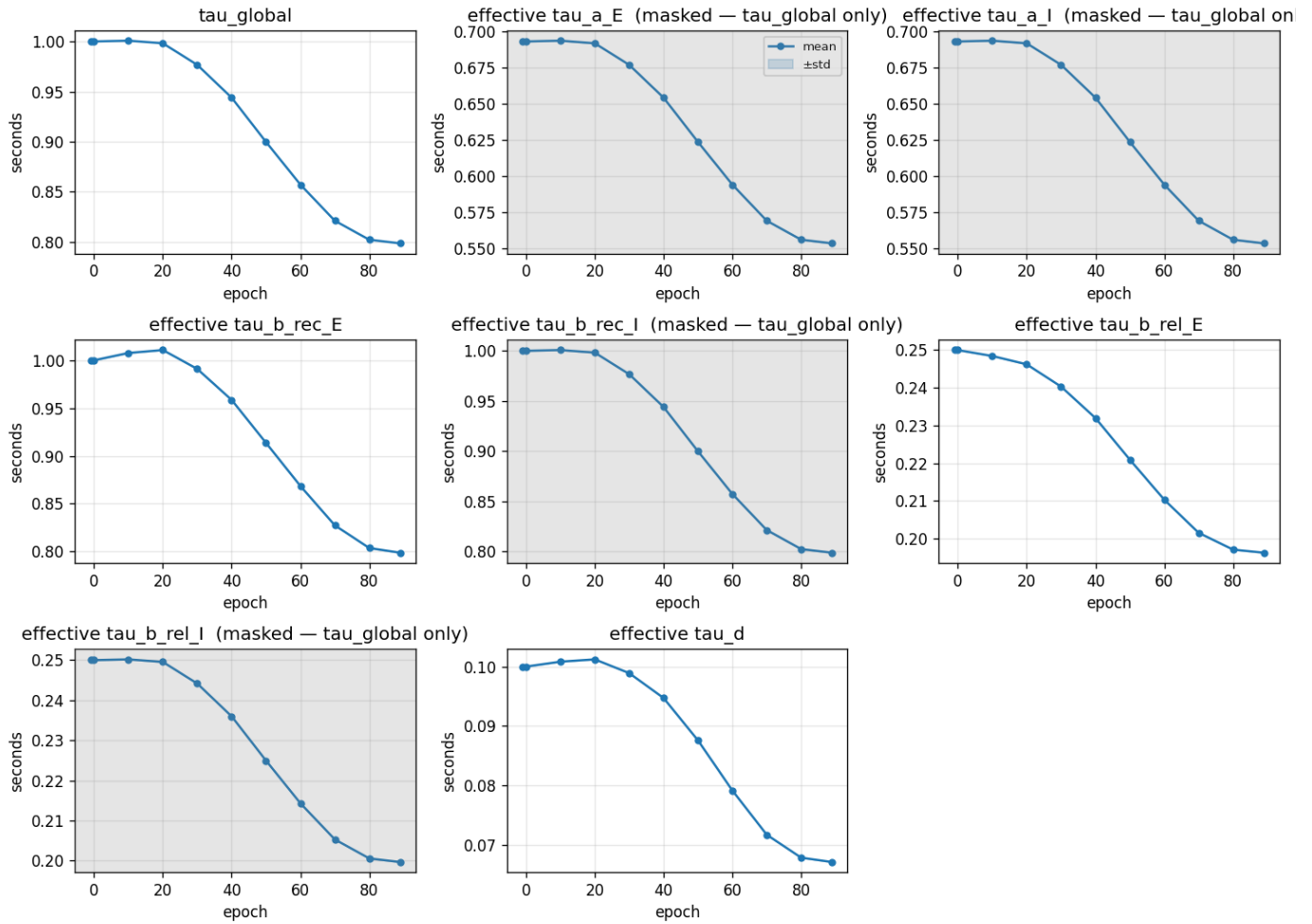
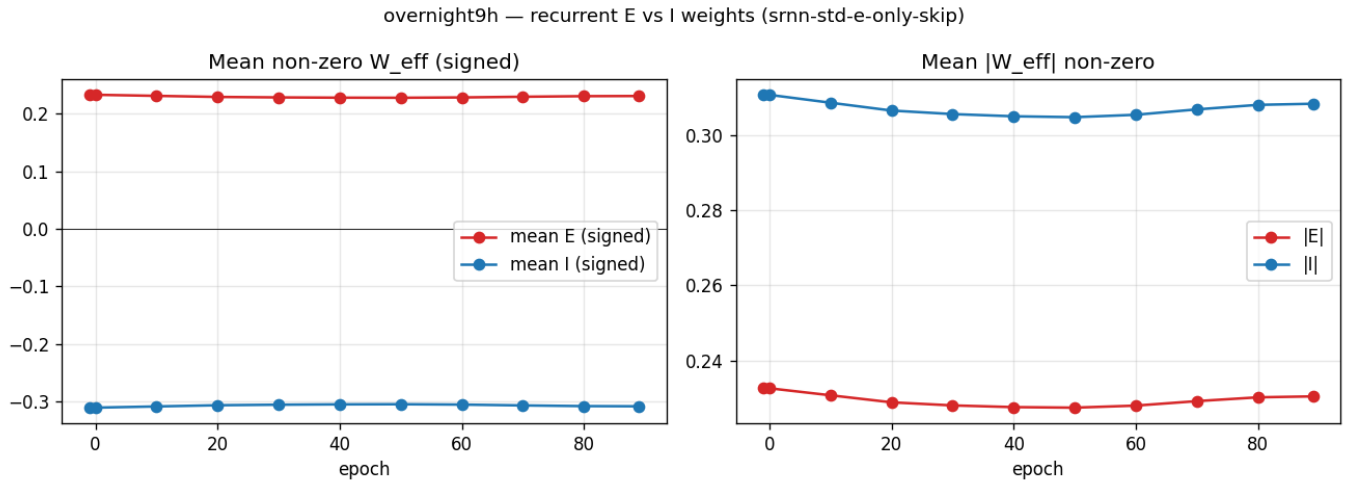


Figure 16: Tau evolution

Figure 17: W_{EI} evolution

srnn-std-e-only-skip — parameter table

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# overnight9h - variant srnn-std-e-only-skip (k=7)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
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effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.79855	-	-	scalar
tau_d (s)	+0.1	+0.067139	0	0	300
tau_a_E (s)	+0.69315	+0.55352	0	0	450
tau_a_I (s)	+0.69315	+0.55352	0	0	450
tau_b_rec_E (s)	+1	+0.7986	0	0	150
tau_b_rel_E (s)	+0.25	+0.19626	0	0	150
tau_b_rec_I (s)	+1	+0.79855	0	0	150
tau_b_rel_I (s)	+0.25	+0.19964	0	0	150
W_{eff} (E src, signed)	+0.23267	+0.2305	0.07641	0.07589	15048
W_{eff} (I src, signed)	-0.31062	-0.30824	0.07724	0.07693	14977
$ W_{\text{eff}} $ (E src)	+0.23267	+0.2305	0.07641	0.07589	15048
$ W_{\text{eff}} $ (I src)	+0.31062	+0.30824	0.07724	0.07693	14977
$W_{\text{in_eff}}$ (input neurons)	+0.0012099	+0.0048252	0.09923	0.1096	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
c_I (SFA coupling)	+0.05	+0.05	0	0	450
c_0_I (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.37122	0	0	300
readout_weight	+0.00034909	-0.0058602	0.06671	0.06043	6675
readout_bias	+0	+0.012499	0	0.01634	89
ic.ic (per-variant init)	+0.15334	+0.16026	0.4128	0.4104	1500